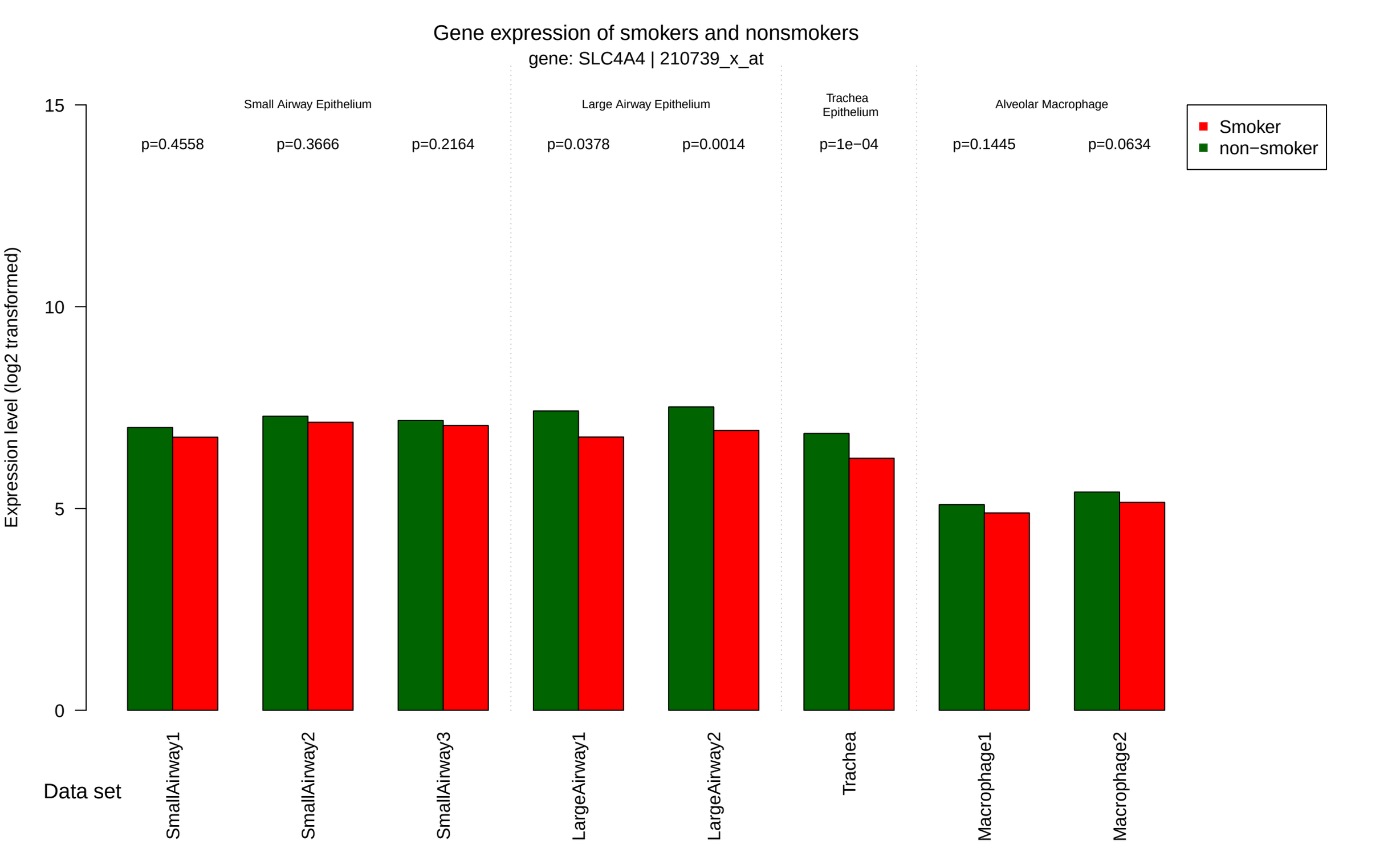
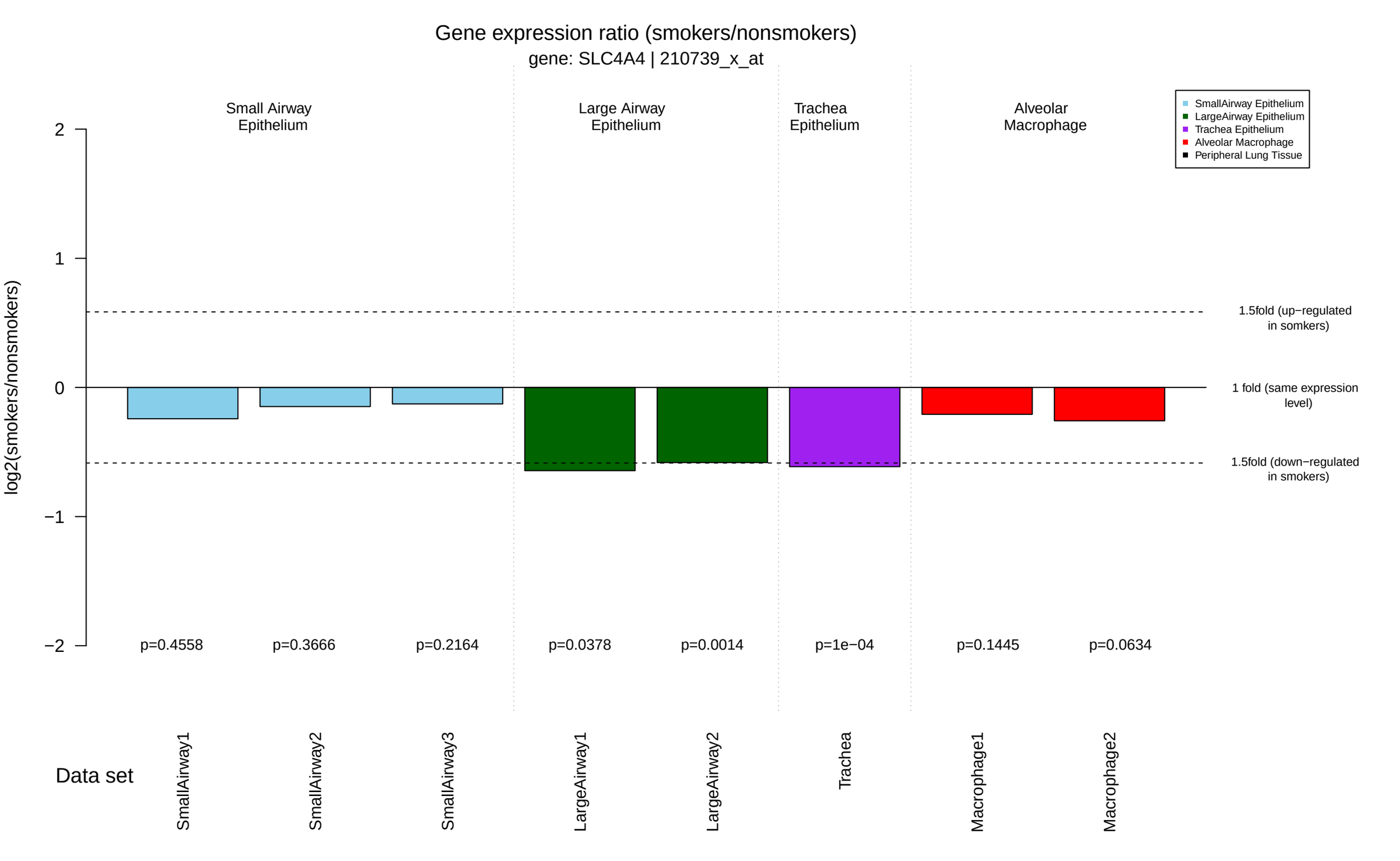
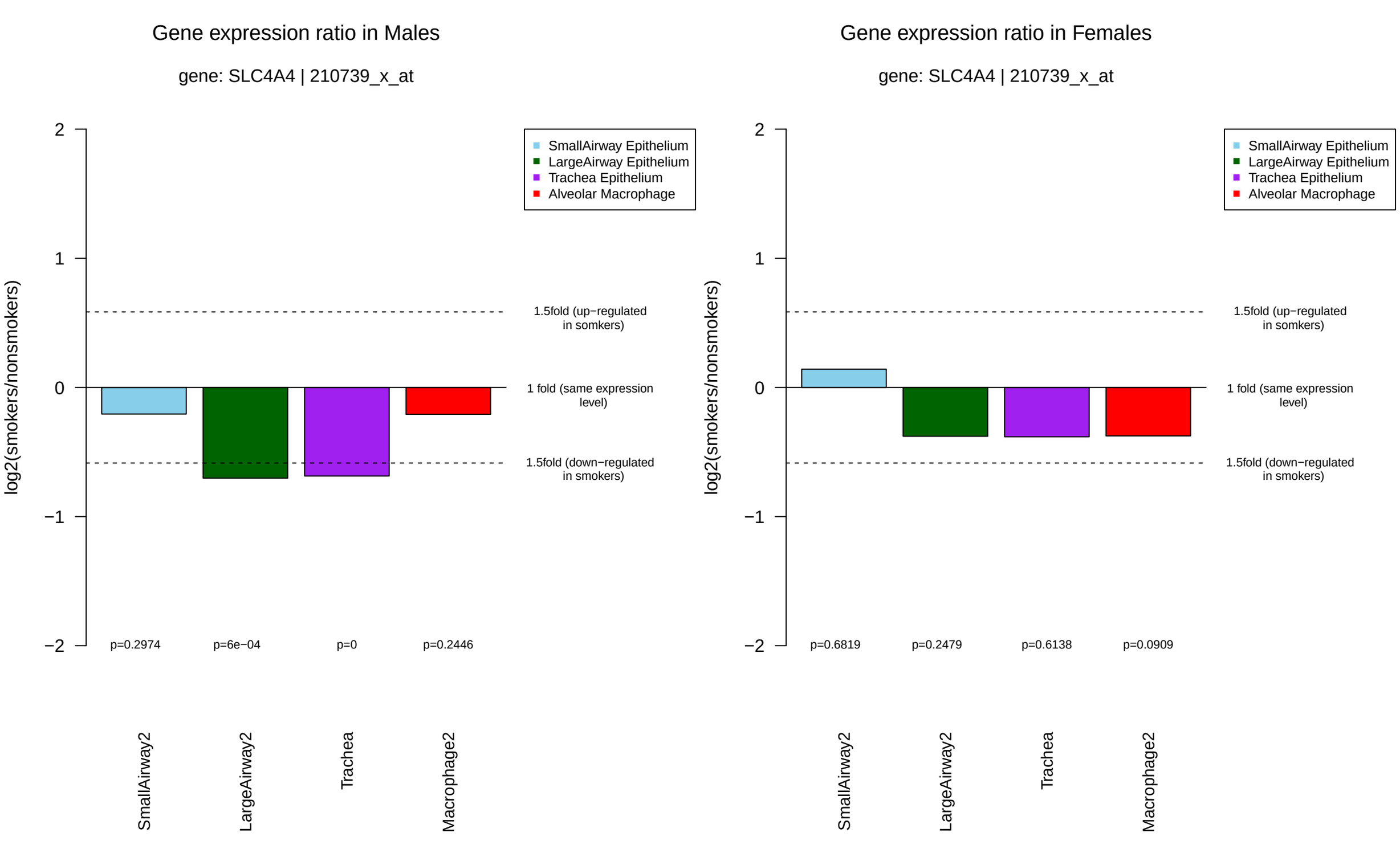
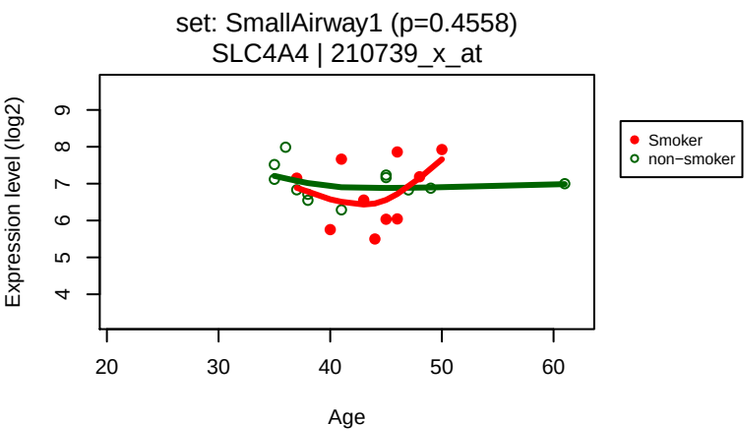
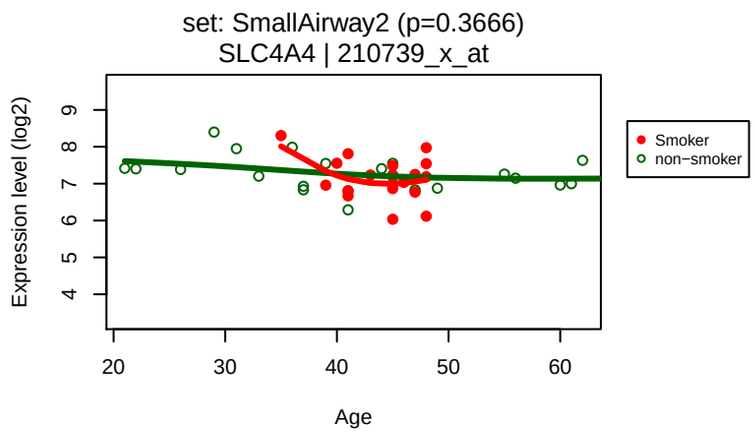
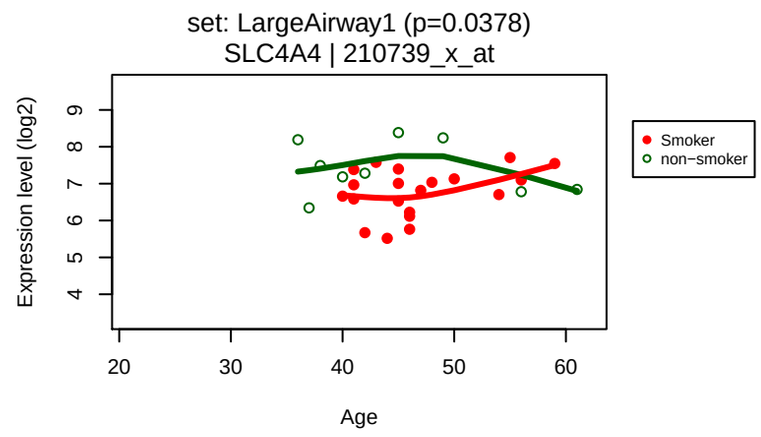

Gene expression of smokers and nonsmokers
gene: SLC4A4 | 210739_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4558
p=0.3666
p=0.2164
p=0.0378
p=0.0014
p=1e−04
p=0.1445
p=0.0634
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

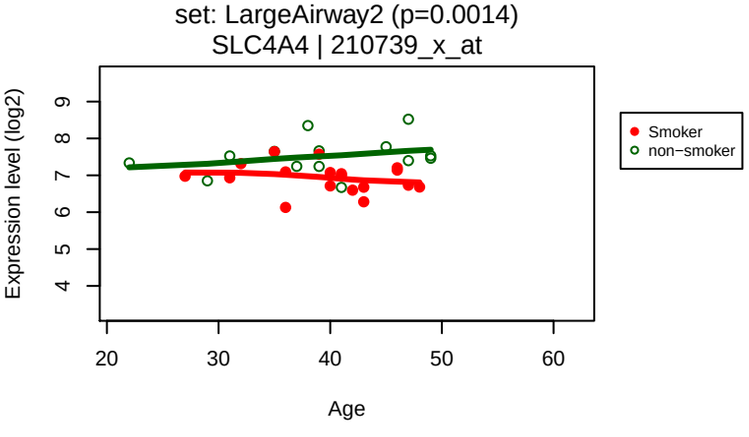

Gene expression ratio (smokers/nonsmokers)
gene: SLC4A4 | 210739_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4558
p=0.3666
p=0.2164
p=0.0378
p=0.0014
p=1e−04
p=0.1445
p=0.0634
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

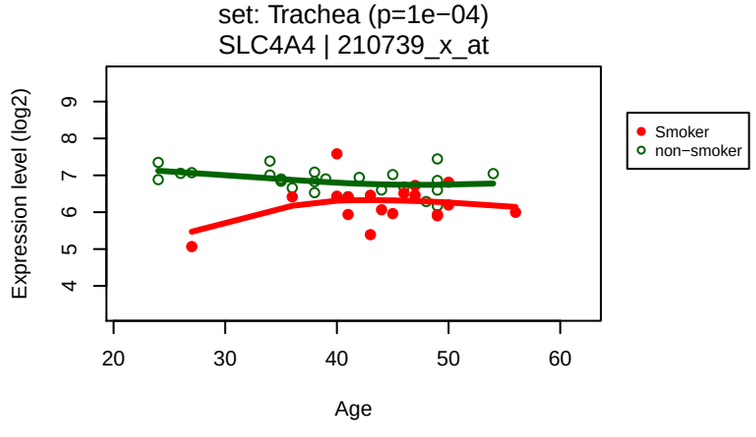

Gene expression ratio in Males
gene: SLC4A4 | 210739_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2974
p=6e−04
p=0
p=0.2446
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC4A4 | 210739_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6819
p=0.2479
p=0.6138
p=0.0909
SmallAirway2
LargeAirway2
Trachea
Macrophage2

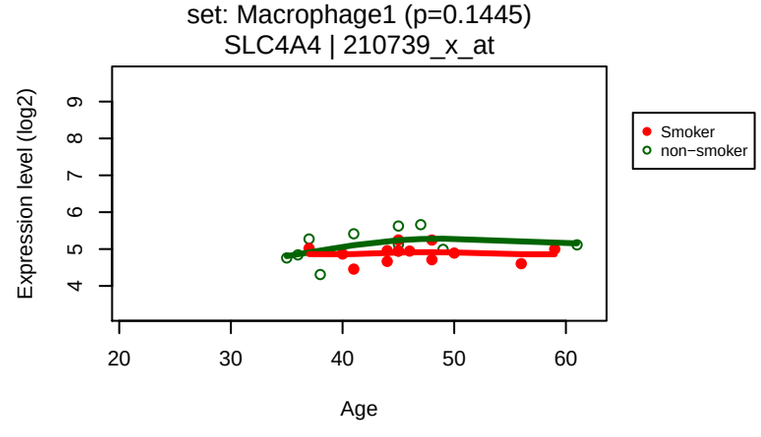

set: SmallAirway1 (p=0.4558)
SLC4A4 | 210739_x_at
Expression level (log2)
Age
Smoker
non−smoker

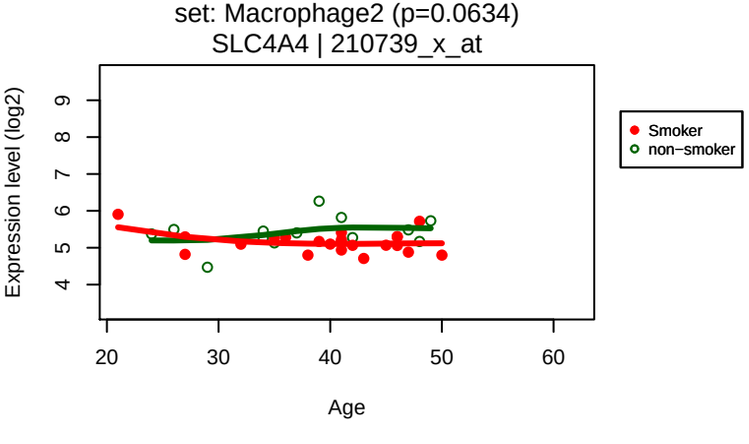

set: SmallAirway2 (p=0.3666)
SLC4A4 | 210739_x_at
Expression level (log2)
Age
Smoker
non−smoker

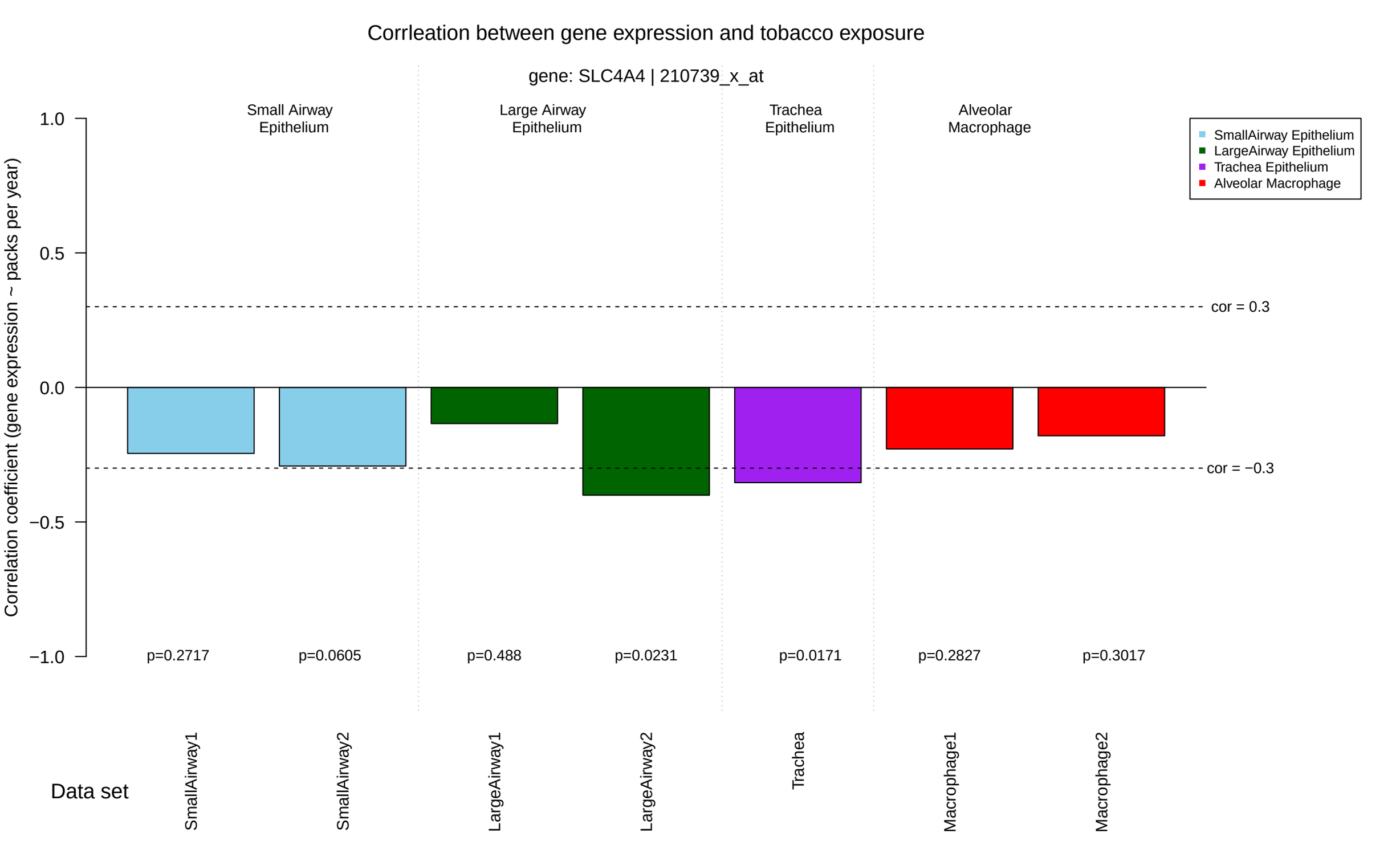

set: LargeAirway1 (p=0.0378)
SLC4A4 | 210739_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0014)
SLC4A4 | 210739_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
SLC4A4 | 210739_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1445)
SLC4A4 | 210739_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0634)
SLC4A4 | 210739_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC4A4 | 210739_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2717
p=0.0605
p=0.488
p=0.0231
p=0.0171
p=0.2827
p=0.3017
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2